

SIMATS ENGINEERING
ASSESSMENT TOOL 2

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COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: Measure network performance using key metrics with simulation tools.
(BL4, BL5)

Assessment Tool Weightage: 30%

Annexure A

Pseudocode Writing Task (Algorithm Design)

1. In a network simulation using Cisco Packet Tracer, a student records total data received as 8000 bits in 4 seconds and observes that 200 packets were sent while 180 packets were received. Write a pseudocode algorithm to calculate the network throughput in bits per second and the packet loss percentage based on the given values. The algorithm should include validation to ensure that time and total packets sent are not zero to avoid errors. It should display both the throughput and packet loss results clearly. Add logic to classify the network as Efficient if throughput is greater than 1500 bps and packet loss is less than 10%, otherwise classify it as Inefficient. Finally, ensure the algorithm outputs a complete summary of the network performance based on the calculated metrics.
2. An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to detect when bandwidth utilization exceeds 80% and recommend switching traffic to a backup link. Explain how this algorithm supports efficient bandwidth management and prevents network congestion.
3. A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay, and utilization values, which change throughout the day. The network administrator wants to continuously identify the best communication path between the headquarters and each

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branch based on overall link quality rather than just the shortest distance. Write a pseudocode algorithm that periodically collects the bandwidth, delay, and utilization of every network link and stores the values in suitable arrays or lists. Use a loop to repeat the monitoring process at regular intervals. Calculate a link quality score for each path using the collected metrics and select the path with the highest score. Include conditions to detect when a link becomes overloaded or fails, automatically selecting an alternate path and generating an alert for the administrator. Explain how your algorithm improves network reliability, routing efficiency, and fault tolerance while adapting to changing network conditions.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
A Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
B Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
C Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
D Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
E Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

	A	B	C	D	E	
Q ₁	2	2	2	1	2	9
Q ₂	2	3	1.5	1	2	9.5
Q ₃	2	2	1.5	0.5	1.5	7.5

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Network Throughput and Packet Loss Calculation:

Pseudocode

BEGIN

SET totalDataReceived = 8000

SET time = 4

SET packetsSent = 200

SET packetsReceived = 180

IF time <= 0 OR packetsSent <= 0 THEN

```
    DISPLAY "Invalid input: Time and packets sent must be greater than
zero."
```

```
ELSE
```

```
    throughput = totalDataReceived / time
```

```
    packetLoss = ((packetsSent - packetsReceived) / packetsSent) * 100
```

```
    DISPLAY "Network Throughput = ", throughput, " bps"
```

```
    DISPLAY "Packet Loss = ", packetLoss, "%"
```

```
    IF throughput > 1500 AND packetLoss < 10 THEN
```

```
        DISPLAY "Network Status: Efficient"
```

```
    ELSE
```

```
        DISPLAY "Network Status: Inefficient"
```

```
    END IF
```

```
    DISPLAY "----- Network Performance Summary -----"
```

```
    DISPLAY "Total Data Received: ", totalDataReceived, " bits"
```

```
    DISPLAY "Time Taken: ", time, " seconds"
```

```
    DISPLAY "Packets Sent: ", packetsSent
```

```
    DISPLAY "Packets Received: ", packetsReceived
```

```
    DISPLAY "Throughput: ", throughput, " bps"
```

```
    DISPLAY "Packet Loss: ", packetLoss, "%"
```

```
END IF
```

```
END
```

Output

- Throughput = $8000 / 4 = \mathbf{2000\ bps}$

- Packet Loss = $((200 - 180) / 200) \times 100 = \mathbf{10\%}$
- Network Status = **Inefficient**

The network is classified as inefficient because, although the throughput is greater than 1500 bps, the packet loss is **not less than 10%**.

2. An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to detect when bandwidth utilization exceeds 80% and recommend switching traffic to a backup link. Explain how this algorithm supports efficient bandwidth management and prevents network congestion.

Bandwidth Utilization Monitoring Algorithm

Pseudocode

BEGIN

INPUT numberOfLinks

CREATE ARRAY bandwidthUsage[numberOfLinks]

WHILE TRUE

FOR i = 1 TO numberOfLinks

INPUT bandwidthUsage[i]

DISPLAY "Link ", i,
" Bandwidth Utilization = ",
bandwidthUsage[i], "% "

IF bandwidthUsage[i] > 80 THEN

DISPLAY "Warning: Link ", i,
" is overloaded."

```

        DISPLAY "Recommendation: Switch traffic to backup link."

    ELSE

        DISPLAY "Link ", i,
            " is operating normally."

    END IF

END FOR

WAIT for a fixed monitoring interval

END WHILE

END

```

Explanation

This algorithm continuously monitors the bandwidth utilization of every network link. The utilization values are stored in an array, and a loop checks each link regularly.

If the bandwidth utilization exceeds **80%**, the link is considered overloaded. The algorithm recommends switching some traffic to a backup link.

This helps in:

- Identifying overloaded links
- Reducing network congestion
- Improving bandwidth utilization
- Maintaining better network performance
- Using backup links when required

Therefore, the algorithm supports efficient bandwidth management and helps prevent network congestion.

3. A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay, and utilization values, which change throughout the day. The network administrator wants to continuously identify

the best communication path between the headquarters and each branch based on overall link quality rather than just the shortest distance. Write a pseudocode algorithm that periodically collects the bandwidth, delay, and utilization of every network link and stores the values in suitable arrays or lists. Use a loop to repeat the monitoring process at regular intervals. Calculate a link quality score for each path using the collected metrics and select the path with the highest score. Include conditions to detect when a link becomes overloaded or fails, automatically selecting an alternate path and generating an alert for the administrator. Explain how your algorithm improves network reliability, routing efficiency, and fault tolerance while adapting to changing network conditions.

Best Path Selection Based on Link Quality

Pseudocode

BEGIN

BEGIN

INPUT numberOfPaths

CREATE ARRAY bandwidth[numberOfPaths]

CREATE ARRAY delay[numberOfPaths]

CREATE ARRAY utilization[numberOfPaths]

CREATE ARRAY qualityScore[numberOfPaths]

CREATE ARRAY linkStatus[numberOfPaths]

WHILE TRUE

FOR i = 1 TO numberOfPaths

INPUT bandwidth[i]

INPUT delay[i]

INPUT utilization[i]

INPUT linkStatus[i]

IF linkStatus[i] = "FAILED" THEN

```

    qualityScore[i] = 0

    DISPLAY "Alert: Path ", i, " has failed."

ELSE IF utilization[i] > 80 THEN

    qualityScore[i] = 0

    DISPLAY "Warning: Path ", i, " is overloaded."

ELSE

    qualityScore[i] =
    bandwidth[i] - delay[i] - utilization[i]

END IF

END FOR

SET bestPath = 1

FOR i = 2 TO numberOfPaths

    IF qualityScore[i] > qualityScore[bestPath] THEN

        bestPath = i

    END IF

END FOR

IF qualityScore[bestPath] > 0 THEN

    DISPLAY "Best Path Selected: ", bestPath
    DISPLAY "Highest Quality Score: ",
        qualityScore[bestPath]

```


ELSE

DISPLAY "No active path is available."

DISPLAY "Alert administrator."

END IF

WAIT for a fixed monitoring interval

END WHILE

END

Explanation

This algorithm continuously collects the **bandwidth, delay, utilization, and status** of every network link. A link quality score is calculated for each path.

- Higher bandwidth improves the quality score.
- Higher delay reduces the quality score.
- Higher utilization reduces the quality score.
- A failed or overloaded link is avoided.

The algorithm selects the path with the **highest quality score**. If the current path becomes overloaded or fails, another available path can be selected automatically.

This improves:

- **Network reliability** by avoiding failed links
- **Routing efficiency** by selecting the best available path
- **Fault tolerance** by using alternate paths
- **Network performance** by adapting to changing conditions

Thus, the algorithm provides a dynamic and efficient method for selecting the best communication path between the headquarters and branch offices.

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